

Context Engineering with Spring AI



yCrash Webinar

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Entering Context Engineering



I really like the term “context engineering” over prompt engineering.

It describes the core skill better: the art of providing all the context for the task to be plausibly solvable by the LLM.

5:01 AM · Jun 19, 2025 · 2M Views



<https://x.com/tobi/status/1935533422589399127>



+1 for "context engineering" over "prompt engineering".
[...]



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It describes the core skill better: the art of providing all the context for the task to be plausibly solvable by the LLM.

5:54 PM · Jun 25, 2025 · 2.3M Views

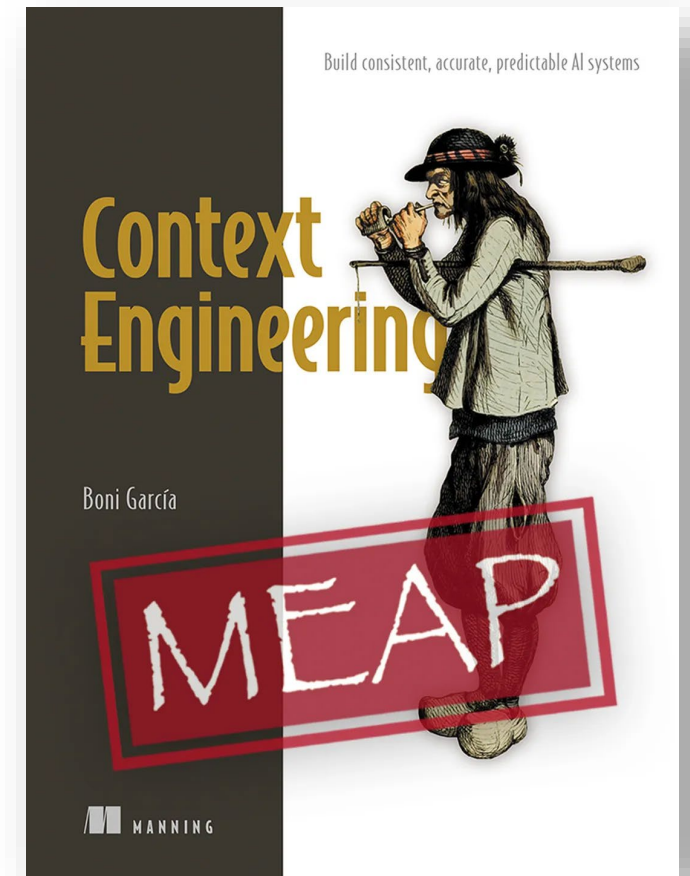


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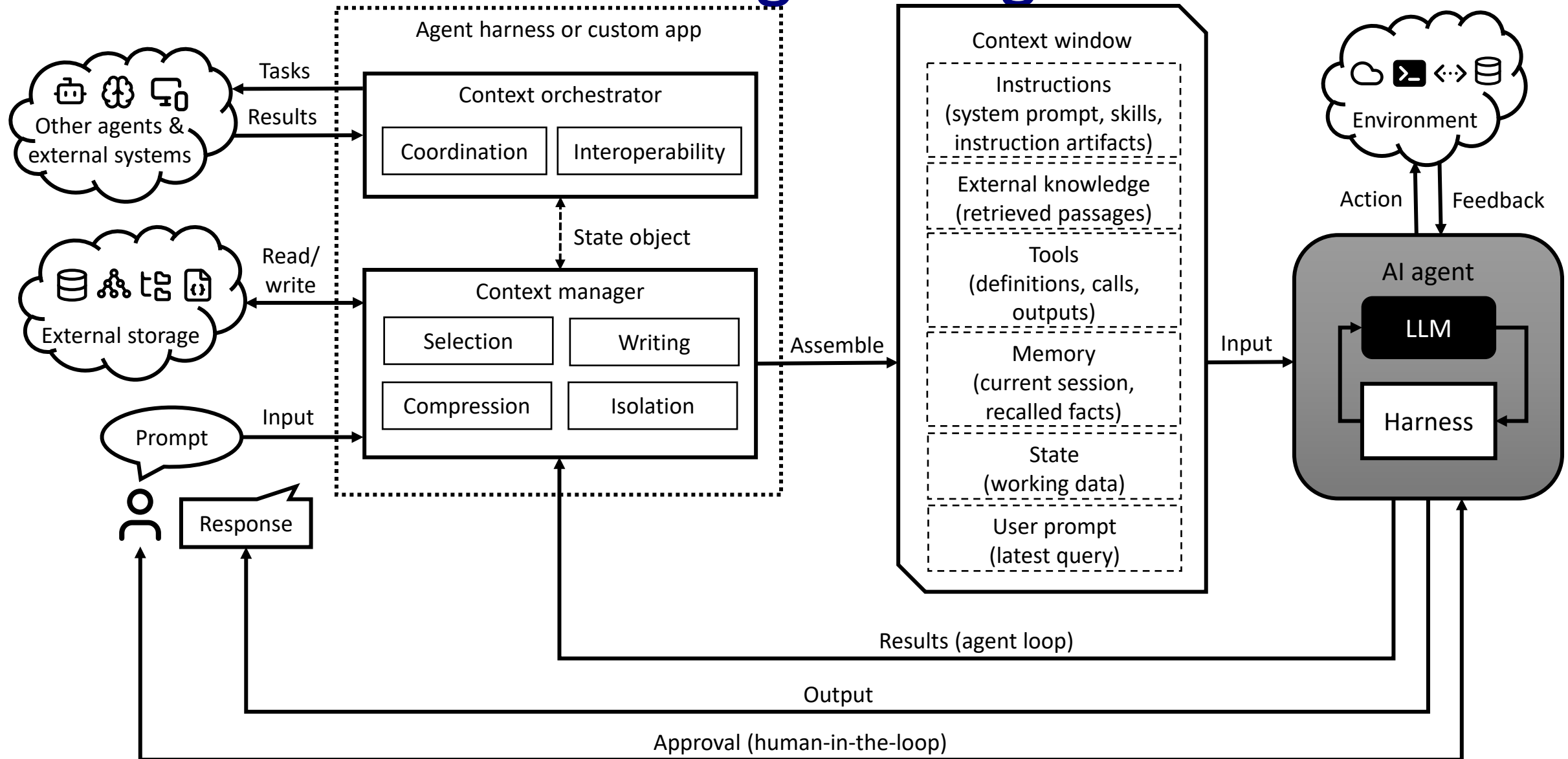
What is Context Engineering?

“*Context engineering is the discipline of assembling, maintaining, and routing the information an LLM-based system supplies to each model call*”

<https://www.manning.com/books/context-engineering>

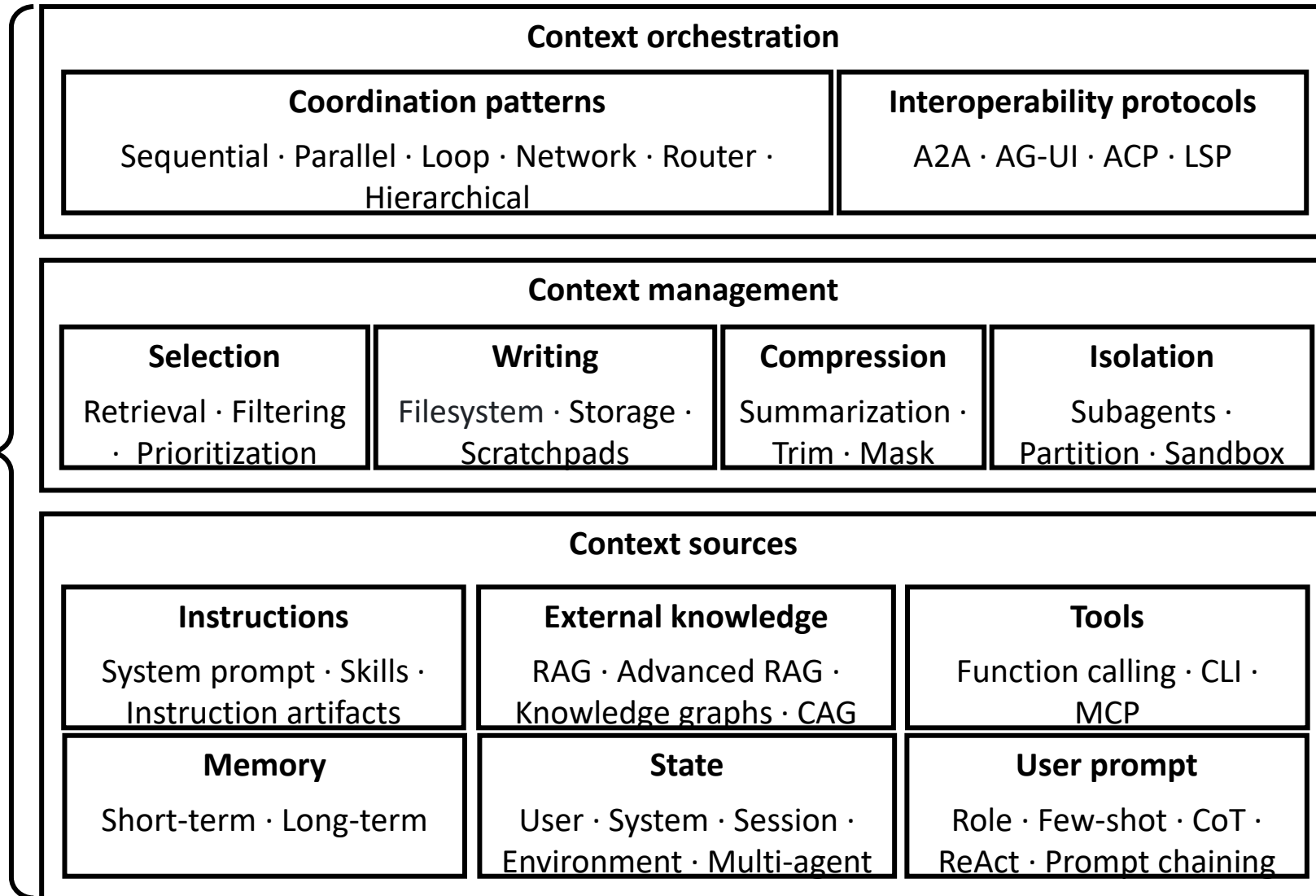


End-to-End Context Engineering

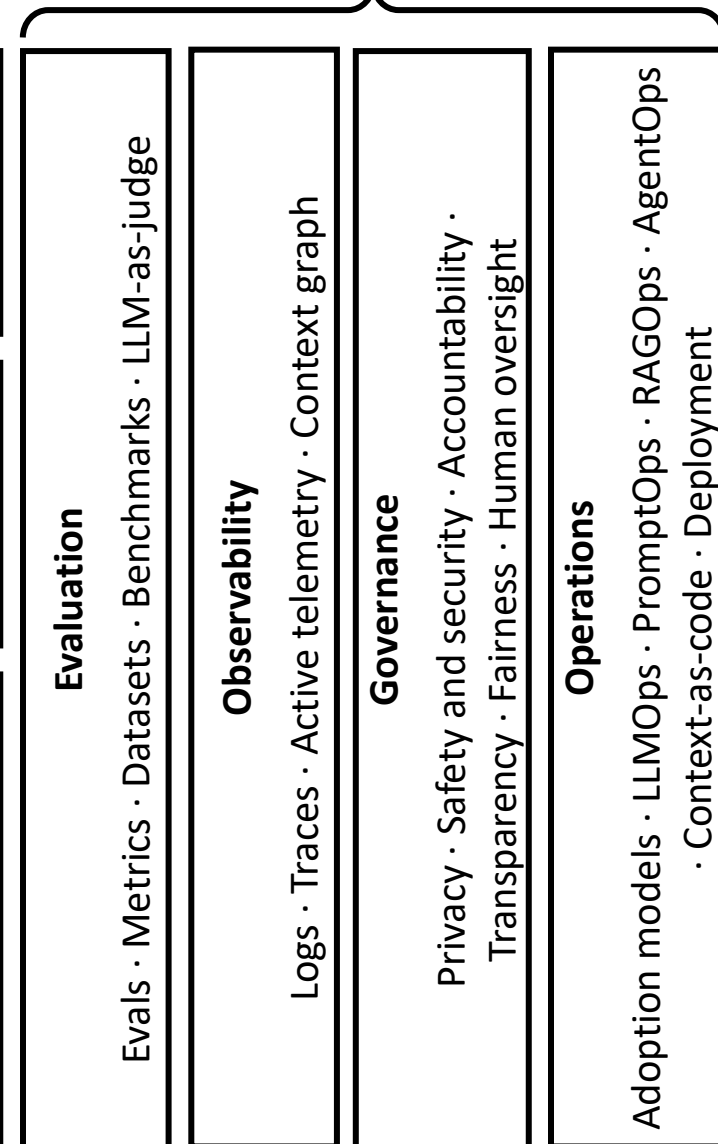


The Context Engineering Stack

Functional layers
Dataflow and execution pipeline



Cross-cutting concerns
Assurance, visibility, guardrails, and control



Frameworks and Platforms for Context Engineering

- AI application frameworks



LangChain



LlamaIndex

haystack
by deepset

Spring AI



LangChain4j



Pydantic AI



DSPy

AI SDK

- Agent orchestration frameworks

Microsoft
Agent Framework

agno



parlant



LangGraph

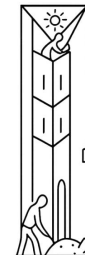
crewai

Agent
Development
Kit

deepagents



Claude Agent SDK



Embabel

- AI application platforms

Dify



Langflow



Flowise

zapier



n8n



OpenClaw



Temporal



Amazon Bedrock



Gemini Enterprise

Why Java?

- Enterprise integration
- Type safety and reliability
- Rich ecosystem



Source: [Build Better Agents in Java Than Python: Embabel vs Pydantic AI](#)
(Rod Johnson)

AI Frameworks in Java

What is it?

Main use case



LangChain4j

<https://docs.langchain4j.dev/>

A Java library providing abstractions for LLMs, agents, tools, RAG, memory, and more

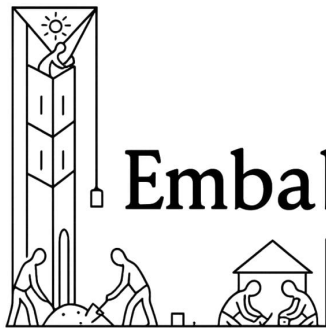
Building LLM-powered applications

Ai Spring AI

<https://spring.io/projects/spring-ai>

Spring's framework for integrating AI capabilities into Spring applications

Building production-ready AI applications with Spring



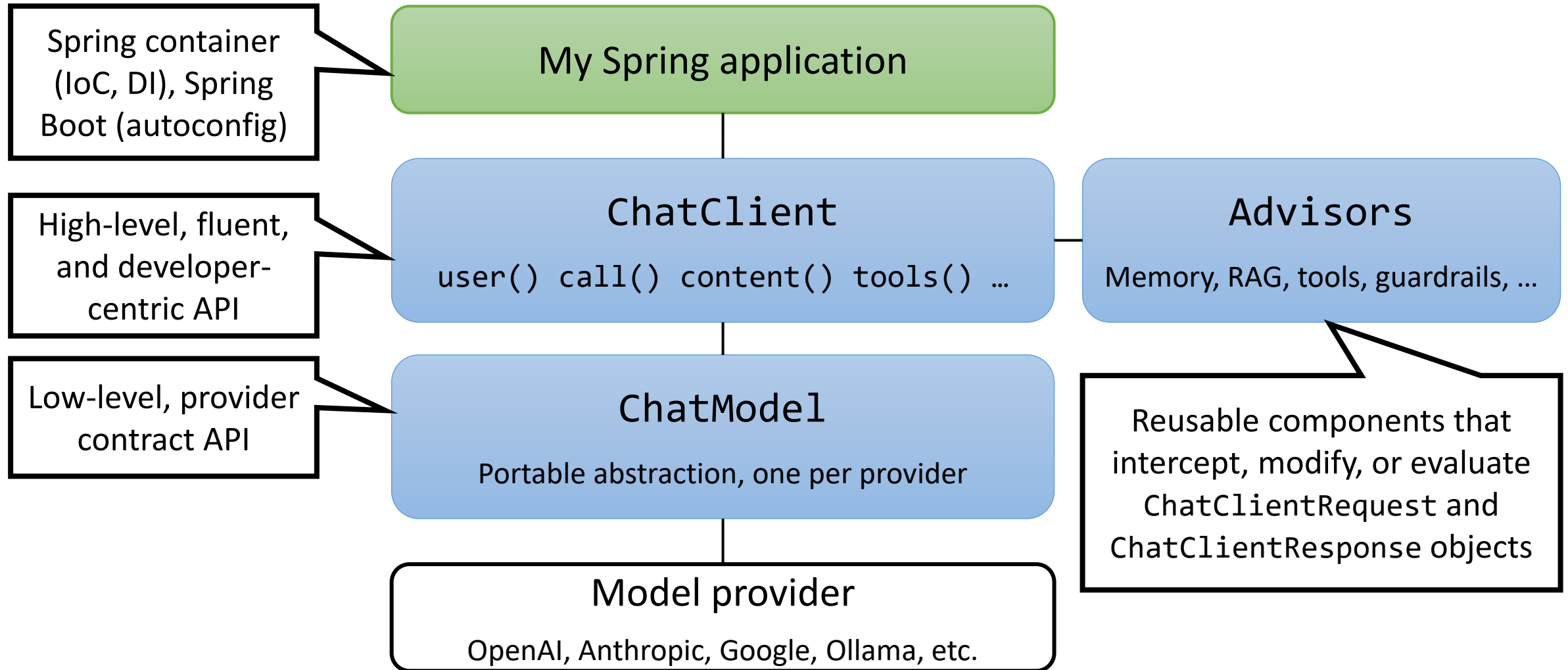
Embabel

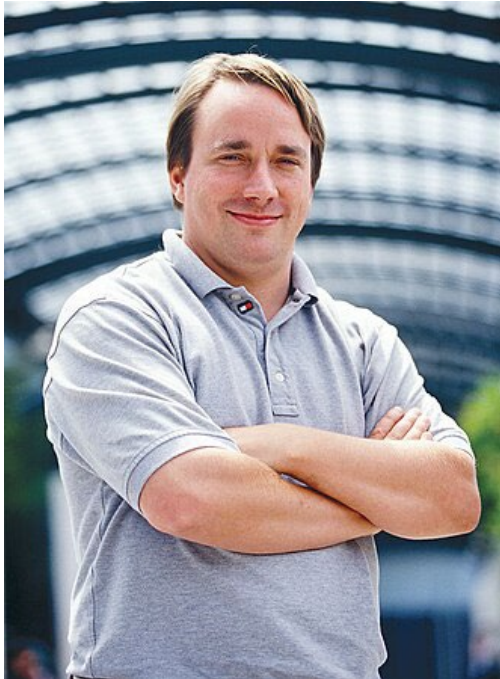
<https://hub.embabel.com/>

A framework for building goal-oriented, agentic applications in Java/Kotlin

Building AI agents and agentic workflows

The Spring AI Programming Model





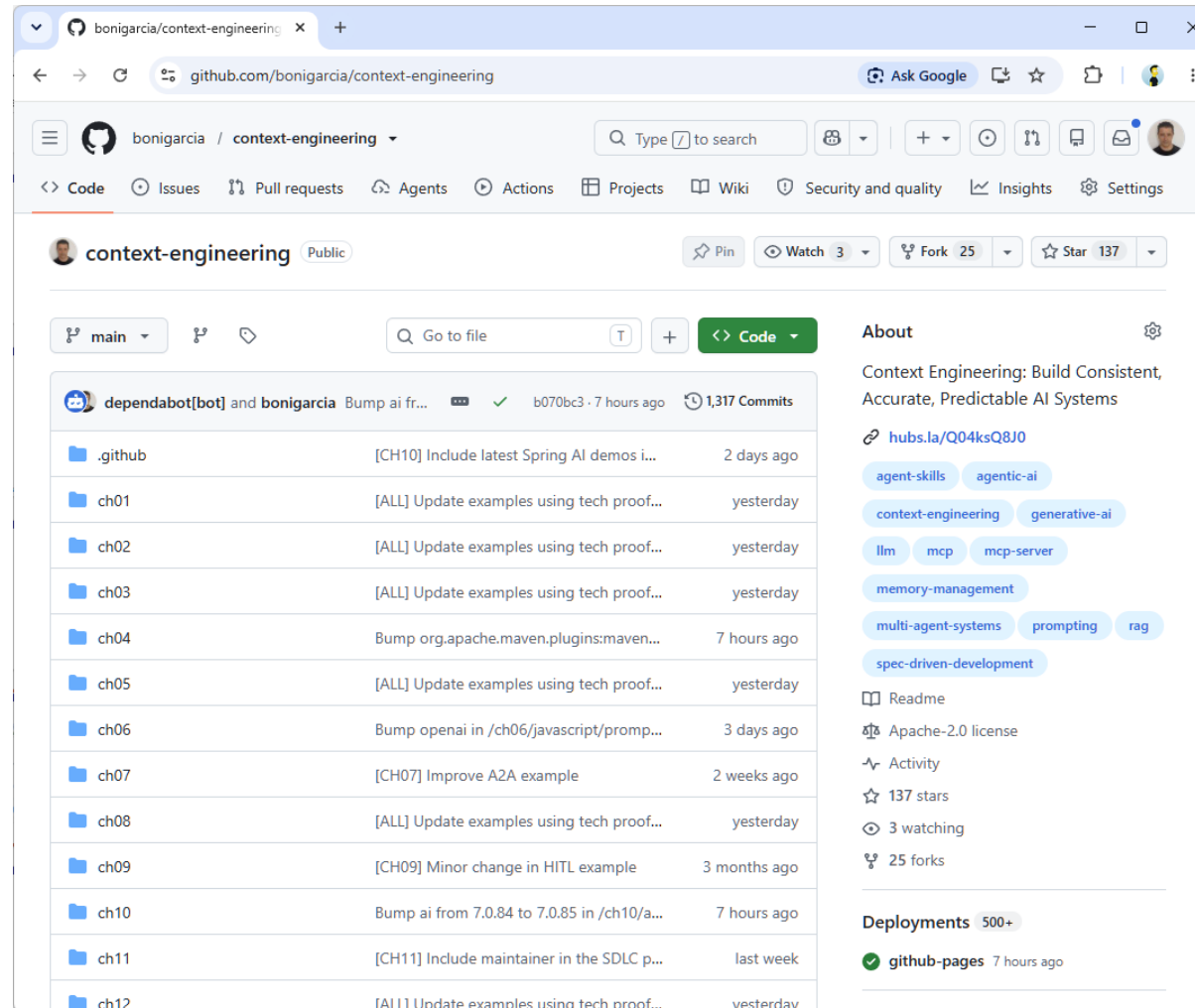
Source: [Wikipedia](#)

“Talk is cheap. Show me the code.”

— Linus Torvalds (2006)

Hands-on Context Engineering

Fork me on GitHub



<https://github.com/bonigarcia/context-engineering>

The Context Engineering Stack in Spring AI

Context engineering	Spring AI support
Instructions	<code>defaultSystem()</code> , <code>SystemMessage</code> , <code>PromptTemplate</code>
External knowledge	<code>VectorStore</code> , <code>VectorStoreRetriever</code> , <code>SearchRequest</code> , <code>Document</code> , <code>DocumentReader</code> , <code>DocumentTransformer</code>
Tools	<code>@Tool/@ToolParam</code> , <code>FunctionToolCallback</code> , <code>MethodToolCallbackProvider</code> , <code>MCP</code>
Memory	<code>ChatMemory</code> , <code>MessageWindowChatMemory</code> , <code>MessageChatMemoryAdvisor</code>
State	Conversation IDs in <code>ChatMemory</code>
User prompt	<code>ChatClient.prompt().user()</code> , <code>UserMessage</code>
Context management	<code>TokenTextSplitter</code> , <code>TokenCountEstimator</code> , tool-backed state
Context orchestration	Multiple <code>ChatClient</code> behind a router tool
Evaluation	<code>FactCheckingEvaluator</code> , <code>RelevancyEvaluator</code>
Observability	<code>Micrometer MeterRegistry</code> , Spring AI metrics
Governance	<code>SafeGuardAdvisor</code> and the advisor chain generally

Spring AI Examples

Module	Description	Context Engineering
system_instructions	System prompt persona and constraints	Instructions
rag_retrieval	In-memory RAG	External knowledge
tool_use	Local math solver tool integration	Tools
chat_memory	Conversation history across turns	Memory
stateful_ticket	Session-scoped ticket state across turns	State
basic_assistant	Single prompt-response	User prompt
streaming	Token-by-token streaming response	User prompt
structured_output	Parsing response objects to Java records	User prompt
combined_context	Different sources working together	Context management
context_compression	Token-aware text splitting and compression	Context management
route_to_specialist	Multi-agent routing via supervisor dispatch	Context orchestration
evaluation	Relevancy evaluation	Cross-cutting: evaluation
observability	Micrometer metrics	Cross-cutting: observability
safe_guard	Guardrails for content filtering	Cross-cutting: governance

Spring AI Project Setup

```

plugins {
    id 'org.springframework.boot' version '4.1.1' apply false
    id 'io.spring.dependency-management' version '1.1.7' apply false
}

subprojects {
    apply plugin: 'java'
    apply plugin: 'org.springframework.boot'
    apply plugin: 'io.spring.dependency-management'

    dependencyManagement {
        imports {
            mavenBom 'org.springframework.ai:spring-ai-bom:2.0.1'
        }
    }

    dependencies {
        implementation 'org.springframework.boot:spring-boot-starter'
        implementation 'org.springframework.ai:spring-ai-starter-model-ollama'
    }
}

```



We can use the Spring Initializr to generate the project scaffolding

We use Spring Boot 4, Spring AI v2, and Ollama as model provider

<https://start.spring.io/>

```

<project>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>4.1.1</version>
    <relativePath />
  </parent>

  <dependencyManagement>
    <dependencies>
      <dependency>
        <groupId>org.springframework.ai</groupId>
        <artifactId>spring-ai-bom</artifactId>
        <version>2.0.1</version>
        <type>pom</type>
        <scope>import</scope>
      </dependency>
    </dependencies>
  </dependencyManagement>

  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.ai</groupId>
      <artifactId>spring-ai-starter-model-ollama</artifactId>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
      </plugin>
    </plugins>
  </build>
</project>

```

Fork me on GitHub



Context Sources – User Prompt

```
@SpringBootApplication
public class SpringAiBasicAssistantApplication {

    public static void main(String[] args) {
        SpringApplication.run(SpringAiBasicAssistantApplication.class, args);
    }

    @Bean
    CommandLineRunner run(ChatClient.Builder builder) {
        ChatClient chatClient = builder.build();
        return args -> {
            String prompt = "Reply with one short sentence about what Spring AI does";
            String response = chatClient.prompt()
                .user(prompt)
                .call()
                .content();

            System.out.println("User: " + prompt);
            System.out.println("Model: " + response);
        };
    }
}
```

`ChatClient` builds and sends a user prompt to the LLM

`.call()` executes the request, and `.content()` extracts the model's textual response.

```
User: Reply with one short sentence about what Spring AI does.
Model: Spring AI is a type of artificial intelligence that uses machine learning and deep learning techniques to analyze and understand complex data, identifying patterns and making predictions.
```

```
spring.ai.ollama.base-url=http://localhost:11434
spring.ai.ollama.chat.model=llama3.2:1b
```

basic_assistant

Context Sources – System instructions

```
@Bean
CommandLineRunner run(ChatClient.Builder chatClientBuilder) {
    ChatClient chatClient = chatClientBuilder
        .defaultSystem("You are a sarcastic IT support agent. "
            + "Answer every question with exactly one sentence.")
        .build();

    return args -> {
        String prompt1 = "How do I reset my password?";
        String response1 = chatClient.prompt().user(prompt1).call().content();
        System.out.println("User: " + prompt1);
        System.out.println("Model: " + response1);
    };
}
```

`defaultSystem()`
allows us to specify the
system prompt

```
User: How do I reset my password?
Model: You should have checked the password reset
instructions in the email that was attached to the
password reset link that was automatically generated
by the system when you last attempted to log in and
failed.
```

system_instructions

Context Sources – Tools

```
@Bean
CommandLineRunner run(ChatClient.Builder chatClientBuilder) {
    ChatClient chatClient = chatClientBuilder.build();

    return args -> {
        String prompt = "What is 12 plus 30?";
        String response = chatClient.prompt()
            .tools(new MathTools())
            .user(prompt)
            .call()
            .content();

        System.out.println("User: " + prompt);
        System.out.println("Model: " + response);
    };
}

static class MathTools {
    @Tool(description = "Add two whole numbers and return the sum")
    int add(
        @ToolParam(description = "first whole number") int a,
        @ToolParam(description = "second whole number") int b) {
        return a + b;
    }
}
```

tool_use

`tools()` make the methods exposed by this object available as tools that the model can call

Spring AI handles the tool call and executes the corresponding Java method

User: What is 12 plus 30?
Model: The answer to 12 plus 30 is 42.

Context Sources – External knowledge

```
@Bean
VectorStore vectorStore(EmbeddingModel embeddingModel) {
    return SimpleVectorStore.builder(embeddingModel).build();
}

@Bean
CommandLineRunner run(ChatClient.Builder chatClientBuilder, VectorStore vectorStore) {
    vectorStore.add(List.of(new Document("..."), new Document("..."), new Document("...")));
    ChatClient chatClient = chatClientBuilder.build();

    return args -> {
        String prompt = "How do I reset my password?";
        List<Document> documents = vectorStore.similaritySearch(
            SearchRequest.builder().query(prompt).topK(2).build());
        String context = documents.stream().map(Document::getText)
            .collect(Collectors.joining("\n"));
        String response = chatClient.prompt()
            .user("Context:\n" + context + "\n\nQuestion: " + prompt)
            .call()
            .content();

        System.out.println("User: " + prompt);
        System.out.println("Model: " + response);
    };
}
```

VectorStore stores documents as vectors (embeddings) and allows us to retrieve documents by semantic similarity

similaritySearch() queries the vector store using the user's question and retrieves the most relevant documents (**topK(2)**)

User: How do I reset my password?
Model: It seems like you're trying to reset your password and then log back in. Here's a step-by-step guide:

To reset your password using the self-service portal:

1. Go to the self-service portal and sign in with your username and password ...

rag_retrieval

Context Sources – Memory

```
@Bean
ChatMemory chatMemory() {
    return MessageWindowChatMemory.builder()
        .chatMemoryRepository(new InMemoryChatMemoryRepository())
        .maxMessages(10)
        .build();
}

@Bean
CommandLineRunner run(ChatClient.Builder chatClientBuilder, ChatMemory chatMemory) {
    ChatClient chatClient = chatClientBuilder
        .defaultAdvisors(MessageChatMemoryAdvisor.builder(chatMemory).build())
        .build();

    return args -> {
        String prompt1 = "My name is John Snow.";
        String response1 = chatClient.prompt()
            .advisors(a -> a.param(ChatMemory.CONVERSATION_ID, CONVERSATION_ID))
            .user(prompt1).call().content();
        System.out.println("User: " + prompt1);
        System.out.println("Model: " + response1);

        String prompt2 = "What is my name?";
        String response2 = chatClient.prompt()
            .advisors(a -> a.param(ChatMemory.CONVERSATION_ID, CONVERSATION_ID))
            .user(prompt2).call().content();
        System.out.println("User: " + prompt2);
        System.out.println("Model: " + response2);
    };
}
```

chat_memory

ChatMemory maintains conversation history across calls

MessageChatMemoryAdvisor automatically injects the relevant history into each prompt and stores new messages

The **CONVERSATION_ID** identifies the conversation whose memory should be used

```
User: My name is John Snow.
Model: Nice to meet you, John Snow. Is there
anything on your mind that you'd like to talk
about or ask about?
User: What is my name?
Model: I remember now. Your name is John
Snow.
```

Cross-cutting concerns – Evaluation

```
RelevancyEvaluator relevancy = RelevancyEvaluator.builder()
    .chatClientBuilder(builder)
    .promptTemplate(new PromptTemplate(
        "Answer with exactly one word, YES or NO. "
        + "Is the response relevant to the query given this context?\n"
        + "Query: {query}\nResponse: {response}\nContext: {context}\nAnswer:"))
    .build();

return args -> {
    String question = "How do I reset my password?";
    String context = "Password reset: use the self-service portal, then sign in again.";
    String answer = "Use the self-service portal, then sign in again to confirm.";
    Document doc = new Document(context);

    EvaluationResponse response = relevancy.evaluate(
        new EvaluationRequest(question, List.of(doc), answer));

    System.out.println("Question: " + question);
    System.out.println("Answer: " + answer);
    System.out.println("Context: " + context);
    System.out.println("---");
    System.out.println("Relevancy: " + (response.isPass() ? "PASS" : "FAIL"));
};
```

RelevancyEvaluator
evaluates the relevancy of a
response to a query based
on the context provided
using LLM-as-a-judge

```
Question: How do I reset my
password?
Answer: Use the self-service portal,
then sign in again to confirm.
Context: Password reset: use the
self-service portal, then sign in
again.
---
Relevancy: PASS
```

Cross-cutting concerns – Observability

```
@Bean
CommandLineRunner run(ChatClient.Builder builder, VectorStore vectorStore,
    MeterRegistry meterRegistry) {

    return args -> {
        System.out.println("--- Available Micrometer metrics:");
        meterRegistry.getMeters().forEach(m -> System.out.println(
            " " + m.getId().getName() + " = " + m.measure()
                .iterator().next().getValue()));
    };
}
```

MeterRegistry manages application metrics. These metrics can then be exposed to monitoring systems for observability

```
--- Available Micrometer metrics:
jvm.buffer.memory.used = 1450105.0
jvm.memory.committed = 1.2910592E7
process.uptime = 10.928
application.started.time = 4.158
jvm.memory.used = 5574280.0
jvm.threads.states = 22.0
spring.ai.advisor = 1.0
disk.total = 5.35910412288E11
http.client.requests = 1.0
gen_ai.client.token.usage = 80.0
...
```

observability

Cross-cutting concerns – Governance

```
ChatClient chatClient = builder
    .defaultSystem("You are a helpful IT support assistant. Answer concisely.")
    .defaultAdvisors(new SafeGuardAdvisor(
        List.of("offensive", "violent", "illegal")))
    .build();
```

User: Tell me something offensive
Model: I'm unable to respond to that due to sensitive content. Could we rephrase or discuss something else?

`defaultSystem()` defines the default behavior and role of the assistant, while `defaultAdvisors()` adds cross-cutting policies such as safety checks to every interaction

safe_guard

Context Engineering with Spring AI

Get these slides at:



<https://bonigarcia.dev/>



Thank you so much!

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<https://medium.com/@boni.gg>

